



Matteo Merler

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🔗 merlerm.github.io [in merlerm](#) [merlerm](#) [Google Scholar](#)

Summary

I am a researcher at the FBK NLP in Trento, Italy. This Fall, I will start my PhD at the Bethge Lab in Tübingen, Germany, as an ELLIS PhD Student and as part of the International Max Planck Research School for Intelligent Systems (IMPRS-IS). I also work with Tom Silver's group for robotics and planning in Princeton, US. Previously, I obtained my MSc degree in Machine Learning, Data Science and Artificial Intelligence from Aalto University in Helsinki, Finland.

I am interested in developing autonomous AI agents that learn from experience and adapt dynamically to new situations. For this, my research is at the intersection of large-scale vision and language models, reinforcement learning, and planning. In particular, I believe world modeling to be a key component for building intelligent agents that can reason about the world and plan their actions accordingly. I work with both simulated games and real-world robotics as environments for agents to learn and adapt in. I am also curious about the connections between AI and cognitive science, and how insights from human cognition can inform the development of more intelligent agents.

Education

M.Sc. [Aalto University](#), Machine Learning, Data Science and Artificial Intelligence Sept 2022 – Sept 2024

- Graduated with honors, with a **final grade of 5.0/5.0** and a **GPA of 5.0/5.0**.
- **Thesis**: [Generating Code World Models with Large Language Models Guided by Monte Carlo Tree Search](#), supervised by Prof. Pekka Marttinen, published at NeurIPS 2024.
- Thesis awarded as one of the three **best MSc theses of 2024** at the School of Science.
- Received the Dean's Scholarship for outstanding academic performance.

B.Sc. [University of Trento](#), Computer Science Sept 2019 – July 2022

- Graduated with a **final grade of 110/110 cum laude (with honors)** and a **GPA of 28.46/30**.
- Thesis: **Route Optimization for Indoor Spaces**, supervised by Prof. Alberto Montresor (in Italian).

H.S. [Liceo Scientifico Leonardo da Vinci](#), Liceo Scientifico (Italian Scientific High School Diploma) Sept 2014 – July 2019

- Graduated with a **final grade of 100/100**.

Experience

Researcher, [Fondazione Bruno Kessler](#) Trento, Italy
Working as a pre-doctoral researcher ("Researcher 4 contract") in the [Natural Language Processing Lab](#) under the supervision of Bernardo Magnini. Jan 2025 – present

- Lead researcher on FBK's side of a joint project on **planning with Vision-Language Models** (with Aalto University), under resubmission.
- Led a second project on **selectively integrating VLM guidance with model-free Reinforcement Learning** based on the agent's own uncertainty, with a preliminary version accepted at the CAIPI workshop at ECAI 2025 and a full paper in preparation.

- Participating on private research for collaborations with companies.
- Contributing to the draft of a **Horizon Europe funding proposal**, leading the lab's team for Language and Reinforcement Learning.
- Active participation to the research group's scientific activities, including monthly meetings and collaborative discussions.

Research Assistant, [Aalto University](#) [↗](#)

Helsinki, Finland

MSc Thesis worker in the [Foundation Models for Language and Reinforcement Learning](#) [↗](#) (Larel) group, supervised by Prof. Pekka Marttinen. Aug 2024 – Sept 2024

- Conducted a research project on **Symbolic Regression with Large Language Models**, presented at the ACL Student Research Workshop.
- Co-led a research project on **Code World Models** with PhD candidate Nicola Dainese, forming the basis of my Master's Thesis, accepted at NeurIPS 2024.
- Started a project on **planning with Vision-Language Models**, later continued by both Aalto University and my current group at FBK.
- Regularly attended lab meetings and participated in broader academic activities, including the Seminar on Large Language Models.

Internship, [ThinkIN](#) [↗](#)

Trento, Italy

Worked as a software developer intern in collaboration with the University of Trento. Feb 2022 – June 2022

- Designed and built a prototype for an indoor navigation system specifically targeted for supermarkets, to be used with the company's own location tracking system.

Publications

[QVal: Cheaply Evaluating Dense Supervision Signals for Long-Horizon LLM Agents](#) [↗](#)

June 2026

Sergio Hernández-Gutiérrez, **Matteo Merler**[†], Ilze Amanda Auzina[†], Joschka Strüber, Ameya Prabhu, Matthias Bethge
arXiv preprint

[DecSelfMask: Leveraging Unlabeled Text via Self-Relevance-Guided Masking for Decoder-Only Classification](#) [↗](#)

June 2026

Pietro Ferrazzi, **Matteo Merler**, Giovanni Bonetta, Alberto Lavelli, Bernardo Magnini
arXiv preprint

[Guiding Reinforcement Learning with Selective Vision-Language Model Supervision](#) [↗](#)

Oct 2025

Matteo Merler, Giovanni Bonetta, Bernardo Magnini

ECAI Workshop on AI-based Planning for Complex Real-World Applications (CAIPI'25)

[ViPlan: A Benchmark for Visual Planning with Symbolic Predicates and Vision-Language Models](#) [↗](#)

May 2025

Matteo Merler[†], Nicola Dainese[†], Minttu Alakuijala, Giovanni Bonetta, Pietro Ferrazzi, Yu Tian, Bernardo Magnini, Pekka Marttinen
arXiv preprint

[Generating Code World Models with Large Language Models Guided by Monte Carlo Tree Search](#) [↗](#)

Dec 2024

Nicola Dainese[†], **Matteo Merler**[†], Minttu Alakuijala, Pekka Marttinen

Advances in Neural Information Processing Systems 37 (NeurIPS 2024)

In-Context Symbolic Regression: Leveraging Large Language Models for Function Discovery [↗](#)

Aug 2024

Matteo Merler[†], Katsiaryna Haitsiukevich[†], Nicola Daineset[†], Pekka Marttinen

Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics, Volume 4: Student Research Workshop (ACL 2024 SRW)

[†] denotes equal contribution

Additional Experience and Awards ---

Master's Thesis Award [↗](#): My Master's Thesis was awarded as one of the three best MSc theses of 2024 at the School of Science.

Volunteer ---

Volunteer, [Libera Associazioni Nomi e Numeri Contro le Mafie](#) [↗](#)

Calabria, Italy

I participated in a volunteering camp experience in southern Italy where we cultivated the fields of mafia-confiscated houses for the Libera association, while also taking part in various activities on the territory and listening to testimonies from victims and policemen.

July 2017 – July 2017

Volunteer, [Associazione Volontaria "Per un Mondo Migliore"](#) [↗](#)

Rijeka, Croatia

I participated in a volunteering camp experience in Rijeka, Croatia where we helped different families and associations in need around the city and in mainland Croatia while living together in a shared home.

Aug 2018 – Aug 2018

Languages ---

Italian: Native | **English:** C2 (Near-native proficiency) | **German:** A2 (Basic)